

LENGHU, China

WMO: 526020

Lat: 38.8283N

Lon: 93.3787E

Elev: 9091

StdP: 10.47

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.2	-5.1	-31.9	1.3	9.9	-27.7	1.6	9.8	32.0	17.3	28.1	16.0	2.3	320	0.703

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.0	82.6	51.8	79.5	50.1	76.7	48.8	54.9	75.3	52.8	72.6	51.0	71.6	10.0	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
46.4	65.7	63.5	43.5	58.8	62.7	40.4	52.1	62.0	27.2	75.6	25.5	72.7	24.2	71.7	61.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.2	25.2	21.8	DB	-14.7	87.8	3.7	3.5	-17.4	90.3	-19.5	92.3	-21.6	94.2	-24.3	96.7	(4)
(5)			WB	-15.8	56.6	3.4	2.1	-18.3	58.2	-20.3	59.4	-22.2	60.6	-24.7	62.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.3	12.5	19.9	29.3	40.9	50.7	59.4	64.7	62.9	52.7	38.2	24.6	14.3		
	DBStd	19.14	5.81	6.96	7.04	7.17	5.61	4.61	5.11	4.92	5.47	6.16	6.48	5.19		
	HDD50	5254	1161	843	642	283	59	2	0	29	366	763	1106			
	HDD65	9483	1626	1263	1107	723	442	174	67	99	369	829	1213	1571		
	CDD50	1348	0	0	0	10	82	285	457	401	111	2	0	0		
	CDD65	102	0	0	0	0	0	7	59	35	1	0	0	0		
	CDH74	1291	0	0	0	1	11	123	689	444	23	0	0	0		
	CDH80	237	0	0	0	0	0	7	153	76	1	0	0	0		
	WSAvg	8.0	5.4	7.6	9.3	9.6	9.6	9.5	9.4	8.8	8.1	7.5	5.9	4.9		
	PrecAvg	0.7	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.1	0.1	0.0	0.0	0.0		
	PrecMax	1.5	0.3	0.1	0.1	0.3	0.4	1.1	0.6	0.5	0.8	0.1	0.0	0.1		
(15) Precipitation	PrecMin	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	0.3	0.0	0.0	0.0	0.0	0.1	0.2	0.1	0.2	0.2	0.0	0.0	0.0		
	DB	36.6	46.2	58.8	69.6	75.4	80.7	88.9	86.1	77.1	64.6	49.4	37.7			
	MCWB	24.1	29.1	36.0	41.9	44.6	49.3	56.1	53.7	48.4	40.0	32.1	25.5			
	2% DB	31.6	41.7	52.9	64.5	71.3	77.2	83.9	81.8	72.6	60.2	45.9	32.8			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	21.0	26.8	33.0	39.4	43.1	47.8	52.6	51.2	46.1	38.0	29.8	22.0			
	5% DB	28.5	37.6	48.7	61.0	68.4	74.9	80.7	78.7	69.4	56.3	42.6	29.6			
	MCWB	19.0	24.6	30.8	37.9	41.9	46.5	51.0	49.6	44.3	36.1	28.1	20.1			
	10% DB	25.1	33.9	45.0	57.3	65.5	72.5	77.7	75.9	66.3	52.8	38.9	26.6			
	MCWB	16.9	22.4	28.9	36.0	40.8	45.8	49.7	48.8	42.7	34.3	26.0	18.2			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	24.1	29.6	36.3	42.4	45.9	53.1	58.1	56.9	51.9	41.0	32.5	25.5			
	MCDB	36.0	45.0	57.9	68.0	71.3	71.2	83.7	77.8	69.2	61.4	48.4	36.7			
	2% WB	21.2	27.1	33.2	40.1	44.2	50.3	55.7	54.7	48.3	38.5	30.1	22.3			
	MCDB	31.3	41.2	51.8	63.4	67.9	69.3	77.2	73.7	67.1	58.4	45.1	32.1			
	5% WB	19.1	24.7	31.0	38.2	42.8	48.7	53.5	52.5	46.0	36.6	28.3	20.3			
(33) Mean Daily Temperature Range	MCDB	28.2	37.4	48.1	59.9	65.9	69.2	74.7	71.0	65.4	55.4	42.4	29.4			
	10% WB	16.9	22.5	29.1	36.4	41.4	47.3	51.6	50.7	44.1	34.7	26.2	18.2			
	MCDB	24.9	33.6	44.7	56.5	63.5	68.4	73.2	71.2	63.0	51.9	38.4	26.4			
	MDBR	27.3	27.8	28.5	28.0	25.7	23.5	23.0	24.2	25.9	28.1	28.1	26.9			
	5% DB	29.5	30.2	32.6	31.2	29.5	27.1	26.5	27.5	28.1	30.4	30.0	28.6			
(40) Clear-Sky Solar Irradiance	MCWBR	21.2	19.7	18.6	15.5	13.7	10.7	9.3	9.9	12.0	15.4	18.0	20.4			
	MCDDBR	28.8	29.5	31.2	29.3	26.0	21.4	21.5	21.1	23.0	28.9	29.3	27.8			
	MCWBR	20.7	19.3	18.0	14.8	12.3	8.8	7.9	8.3	10.1	14.7	17.7	19.8			
	taub	0.249	0.293	0.352	0.416	0.431	0.442	0.429	0.403	0.368	0.310	0.270	0.245			
	taud	2.345	2.191	2.001	1.840	1.861	1.902	2.001	2.059	2.114	2.212	2.298	2.365			
(42) All-Sky Solar Radiation	Ebn at Noon	298	294	284	272	270	266	269	273	276	283	285	292			
	Edh at Noon	28	37	51	64	64	62	55	51	45	36	29	25			
	RadAvg	930	1255	1723	2095	2322	2297	2213	2102	1797	1466	1053	843			
(45)	RadStd	56	49	53	80	85	128	96	87	80	43	38	35			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-239	+110	N/A	(46)	
(47)	Regional (1 neighbor)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	-191	+151	+56	(47)	

Nomenclature: See separate page